

1. Administrative & Study Identification

**Full Study Title:** A Study of MIRCERA in Participants With Chronic Kidney Disease Stage 3-4 and Not on Dialysis **Short Title/Acronym:** MIRCERA in Pre-Dialysis CKD **Protocol Number:** Hoffmann-La Roche Internal Study ID **NCT Number:** NCT02567188 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** MIRCERA (methoxy polyethylene glycol-epoetin beta) **Phase of Development:** Observational / Non-interventional **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective** **Primary Objective:** To collect data regarding the safety and efficacy of MIRCERA in participants with a pre-dialysis situation (Chronic Kidney Disease [CKD] stage 3-4) who are treated in a normal clinical environment. **Secondary Objectives:** To track changes in MIRCERA treatment (dose, frequency, route of administration), evaluate hemoglobin (Hb) fluctuation, monitor initiation of renal replacement therapy, and observe changes in concomitant medications such as iron supplements, immunosuppressives, and antihypertensives.

**2.2 Background & Rationale** Anemia is a major complication for patients with CKD Stage 3-4. This observational study generates real-world evidence regarding the use of MIRCERA in everyday clinical practice. Observing Hb fluctuations and treatment modifications under standard care helps to optimize anemia management protocols and delay adverse cardiorenal complications in pre-dialysis patients.

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3. Study Design & Methodology

**3.1 Design Framework** **Study Type:** Observational, longitudinal, multi-center, non-interventional study **Control Type:** Single-arm (Observational Cohort) **Blinding:** Open-label / Non-blinded **Randomization:** N/A (Observational)

**3.2 Planned Enrollment & Duration** **Target N\$:** 144 participants **Number of Sites:** Multi-center **Estimated Study Start:** 01-Apr-2009 **Estimated Primary Completion:** 01-Apr-2011

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Adult participants above 18 years of age. 2. Pre-dialysis participants with CKD stage 3-4 treated with MIRCERA and continuing in the pre-dialysis state for the whole study period. 3. Estimated glomerular filtration rate (e-GFR) between 15-60 milliliter per minute (mL/min) (calculated).

**4.2 Exclusion Criteria** 1. Participants involved in interventional trials. 2. Participants with a life expectancy less than 1 year or with active malignancy. 3. Planned living kidney transplant within 6 months.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** MIRCERA administered according to the current standard of care and in line with the local summary of product characteristics. **Route of Administration:** Subcutaneous or Intravenous (as determined by routine clinical practice). **Duration of Treatment:** Maximum observation period of 12 months.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard of care therapies including iron supplements, immunosuppressives, and antihypertensive medications. **Prohibited:** Concurrent participation in other interventional clinical trials.

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## 6. Study Timeline & Visit Schedule

*(Note: As a non-interventional study, visits follow standard clinical practice)* | Visit ID | Window | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening/Baseline** | Day 0 | Informed Consent, Medical History, e-GFR assessment, Baseline Hb level, MIRCERA initiation/continuation | | **Interim** | Routine Care | Regular clinical monitoring of Hb, MIRCERA dose adjustments, tracking of concomitant medications | | **End of Study** | Month 12 | Final Hb evaluation, assessment of Hb fluctuation between visits, safety data review |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Hemoglobin (Hb) fluctuation, defined as a change in Hb value > 15 g/L between two consecutive clinical visits from Baseline to Month 12. **Measurement Method:** Standard laboratory blood tests during routine clinical visits.

**7.2 Secondary & Exploratory Endpoints** \* Change in MIRCERA treatment (including dose, frequency, and route of administration). \* Percentage of participants with changes in concomitant iron supplement, immunosuppressive, or

antihypertensive medications. \* Number of participants initiating renal replacement therapy (defined as hemodialysis or peritoneal dialysis).

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Standard real-world pharmacovigilance data collection during the 12-month observational period. **Serious Adverse Events (SAEs):** Evaluated and reported according to standard Hoffmann-La Roche safety reporting timelines. **Data Monitoring Committee (DMC):** Internal safety review by the Sponsor's observational research teams.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Observational Set / Safety Set (all enrolled participants receiving MIRCERA in the real-world setting). **Sample Size Justification:**  $N=144$ , sufficient to capture real-world treatment trends and Hb fluctuations in the target population. **Handling of Missing Data:** Dealt with using standard statistical principles for longitudinal observational data (e.g., descriptive statistics with available data).

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** The study was conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local regulations for observational/non-interventional research. **Monitoring Plan:** Centralized review of electronic case report forms (eCRFs) and source data verification corresponding to routine medical records. **Audit Trail:** Secure database logging of MIRCERA dispensing, laboratory outputs, and documented medical events.

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## 11. Study Identifiers & Governance

**Primary ID:** Hoffmann-La Roche Internal Protocol **Registry Number:** NCT02567188 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** A Study of MIRCERA in Participants With Chronic Kidney Disease Stage 3-4 and Not on Dialysis **Official Regulatory Title:** An observational, longitudinal, multi-center, non-interventional study in participants with chronic kidney disease (CKD) stage 3-4, assessed by Modification of Diet in Renal Disease (MDRD) to collect data regarding safety and efficacy in participants with pre-dialysis situation who are treated with MIRCERA in normal clinical environment.

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## 12. Clinical Framework (Anemia in CKD Specific)

**Primary Condition:** Anemia in Chronic Kidney Disease (CKD Stage 3-4) **Diagnosis Threshold:** e-GFR between 15-60 mL/min **Medical Methodology:** Standard of Care (Normal clinical environment) **Optimization Target:** Minimizing Hemoglobin (Hb) fluctuation to maintain stable levels (fluctuation  $\leq 15$  g/L)

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## 13. Study Procedures & Methodology

**Management Pathway:** Normal clinical environment standard of care **Education Protocol (HCP):** N/A (Non-interventional routine practice) **Education Protocol (Patient):** N/A (Standard clinical counseling for CKD/Anemia) **Quality Audit Mechanism:** Data collection from existing medical records and laboratory values without altering the clinical pathway

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Maximum of 12 months follow-up **Onsite Visit Cadence:** As dictated by standard clinical care appointments over the 12 months **Remote Monitoring Frequency:** Periodic remote extraction of available clinical data **Checklist Review Interval:** Routine clinic intervals

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|------|-----|-----|
| Role                     | Name   | Signature | Date          | :-- | :--  | :-- | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     | Lead |     |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |